

MIT DEEP LEARNING HARDWARE REPORT

Phase-Adaptive Quantization

Design space exploration of 4-bit MMA rescale pipelines

Yichong Zhang and Wenye Xiong

Hardware-aware quantization for custom AI inference accelerators

One-Line Thesis

A single 4-bit quantization pipeline is not globally optimal.

Modern inference has two very different phases:

- **Prefill:** large- M forward pass, compute-bound, rescale arithmetic is expensive.
- **Decode:** $M = 1$ autoregressive generation, memory-bandwidth-bound, 4-bit weight movement helps.

Our search shows that the best datapath changes across workload and phase: C1, C8, and C9 are each selected somewhere under a 0.98 cosine-similarity floor.

Motivation: Same Format, Opposite Behavior

NVFP4 is optimized for general GPUs, but it bundles a fixed datapath:

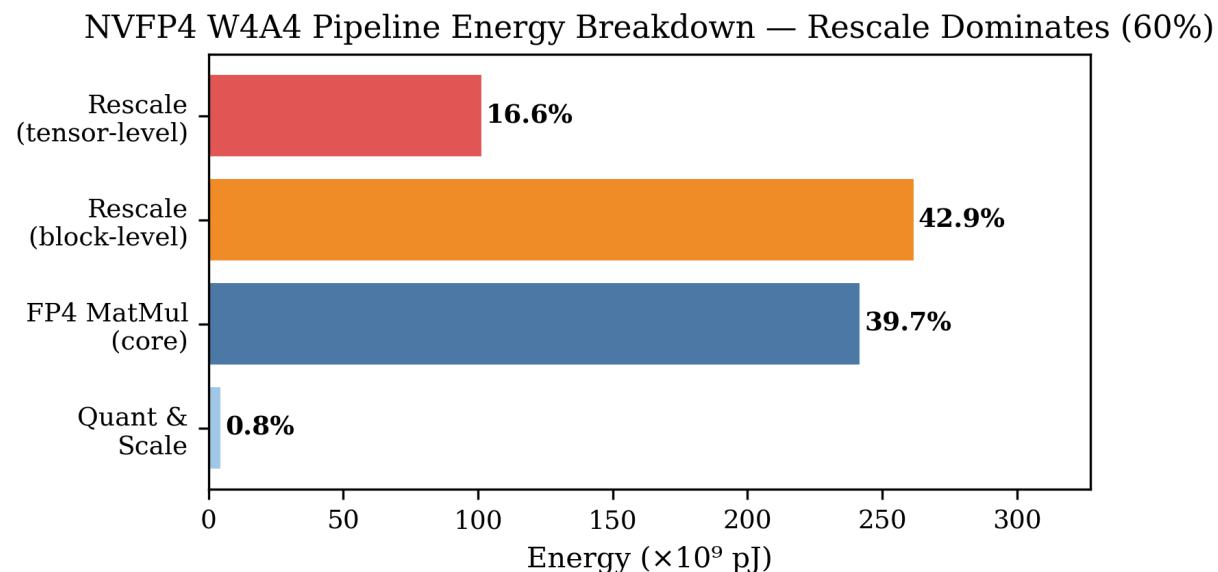
2 rescale levels

b=16 + tensor scale

FP32 rescale

FP32 accumulation

That conservative pipeline is safe, but not always energy efficient for a known deployment target.



Preliminary Bottleneck Result

Phase	NVFP4 energy vs FP16	NVFP4 latency vs FP16	Rescale share	Interpretation
Decode, $M = 1$	0.90x	0.59x	64.5%	Bandwidth savings outweigh rescale cost
Prefill, $M = 128$	1.27x	0.29x	74.0%	Rescale turns W4A4 energy-negative

The same NVFP4-like pipeline helps decode energy but hurts prefill energy. That asymmetry is the reason to search phase-specific datapaths.

Core Research Question

Can we decompose a 4-bit MMA rescale pipeline into hardware-realizable design dimensions and show that **prefill and decode have different Pareto-optimal configurations**?

The design space varies:

- **Pipeline structure:** 0, 1, or 2 rescale levels; block sizes such as 16 or 32.
- **Scale format:** FP32, FP16, or E8M0 power-of-two shifts.
- **Accumulator precision:** FP32 or FP16.
- **Workload shape:** LLM FFN, VLM vision GEMM, VLA action head.

Search Space: Existing Formats Become Points

Config	Role	Topology	Scale path	Accumulator
C0	Ideal lower bound	raw FP4	none	FP32
C1	MXFP4-like	1-level, b=32	E8M0 shift	FP32
C3/C4	One-level baselines	1-level, b=16	FP16 / FP32	FP32
C7	NVFP4-like reference	2-level, b=16 + tensor	FP32 + FP32	FP32
C8	Aggressive one-level	1-level, b=16	FP16	FP16
C9	Aggressive hybrid	2-level, b=16 + tensor	E8M0 + FP16	FP16

Existing schemes are not separate categories; they are fixed coordinates inside the same DSE.

What `run_sweeps.py` Automates

MANIFEST

10 configs x 3
workloads x 2 phases

WORKLOAD YAML

Generate AccelForge
einsum graphs for
each datapath

ARCH YAML

Parametric PE array,
memory, QuantMAC,
FP4MAC,
RescaleMAC

HARDWARE

Run or resume
AccelForge mappings;
record energy,
latency, area

ACCURACY

Pure-Python emulator
computes cosine vs
FP16 snapshots

The proposal path models inference-time execution: weights are prequantized offline, while runtime activation quantization, FP4 matmul, and output rescale remain in the graph.

Experiment Setup

Workload	Representative shape	Phase model	Accuracy proxy
LLM FFN	$(M, 11008, 4096)$	Decode $M = 1$, prefill $M = 128$	Layer-output cosine
VLM vision	$(M, 3072, 3072)$	Decode $M = 1$, prefill $M = 128$	Layer-output cosine
VLA action head	$(M, 256, 4096)$	Decode $M = 1$, prefill $M = 128$	Layer-output cosine

Hardware model: parameterized 8x8 PE-array accelerator, 64 KB GLB, 128 B RF/PE, 45 nm component assumptions.

Completed run status: **60 hardware rows + 60 accuracy rows, all OK.**

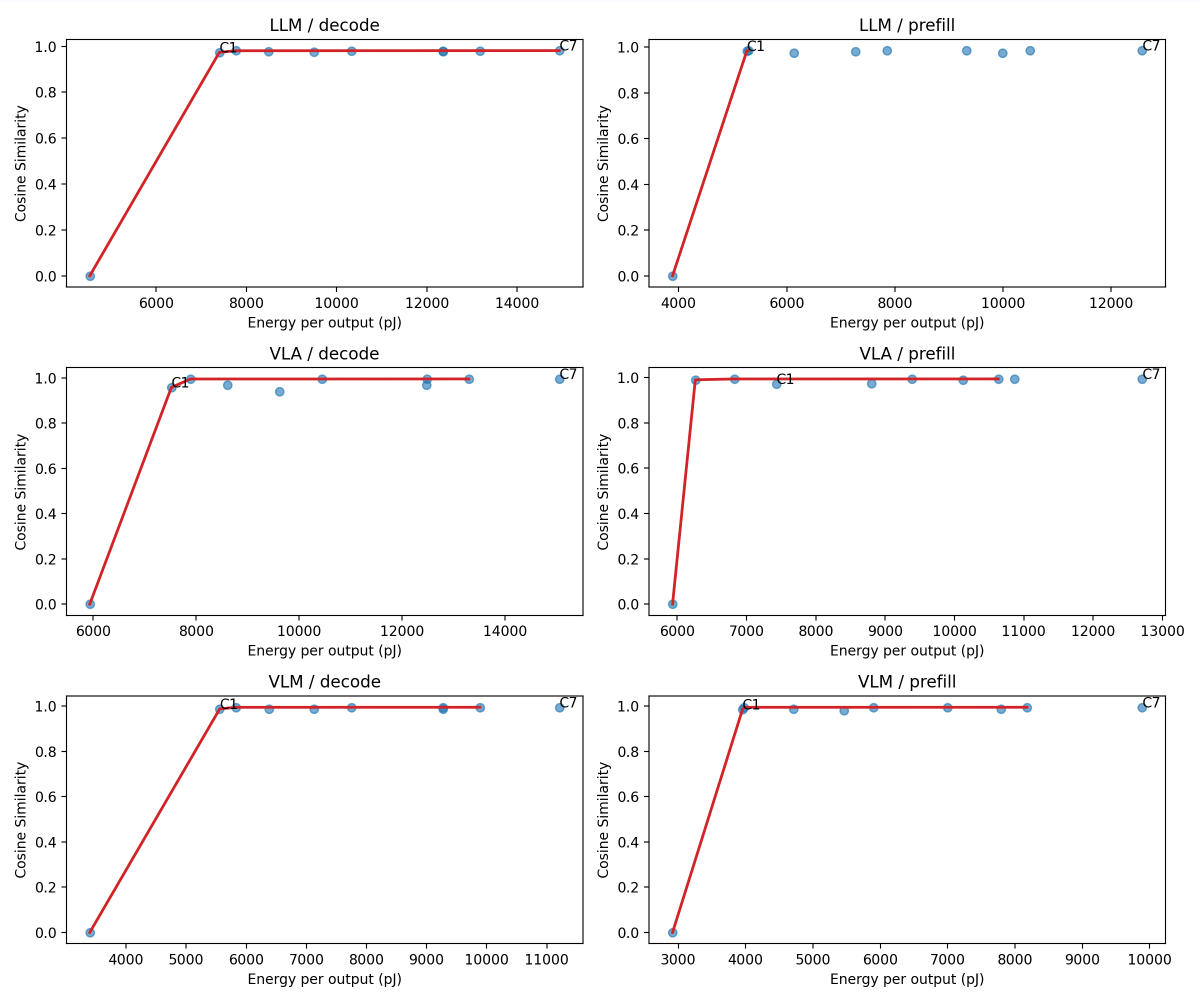
Result 1: Best Configurations Differ

Under cosine similarity ≥ 0.98 :

Workload	Decode best	Decode energy / output	Prefill best	Prefill energy / output
LLM	C8	7,767 pJ	C1	5,262 pJ
VLM	C1	5,557 pJ	C1	3,952 pJ
VLA	C8	7,891 pJ	C9	6,263 pJ

No single configuration wins all six workload-phase cells. VLM collapses to one simple mode; LLM and VLA benefit from phase-specific choices.

Result 2: Pareto Frontiers Shift by Workload and Phase



Each panel plots energy per output against cosine similarity. C7 is NVFP4-like; C1 is MXFP4-like.

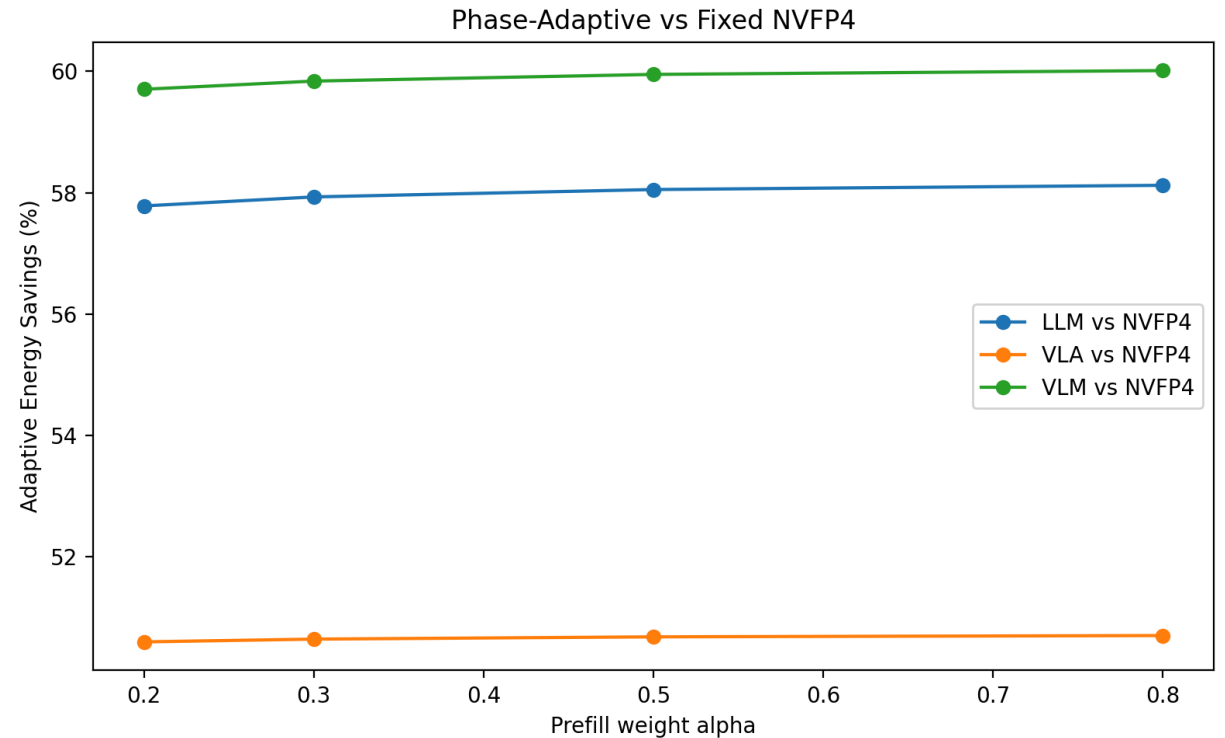
Result 3: Phase-Adaptive Beats Fixed NVFP4-like C7

Across deployment weights α :

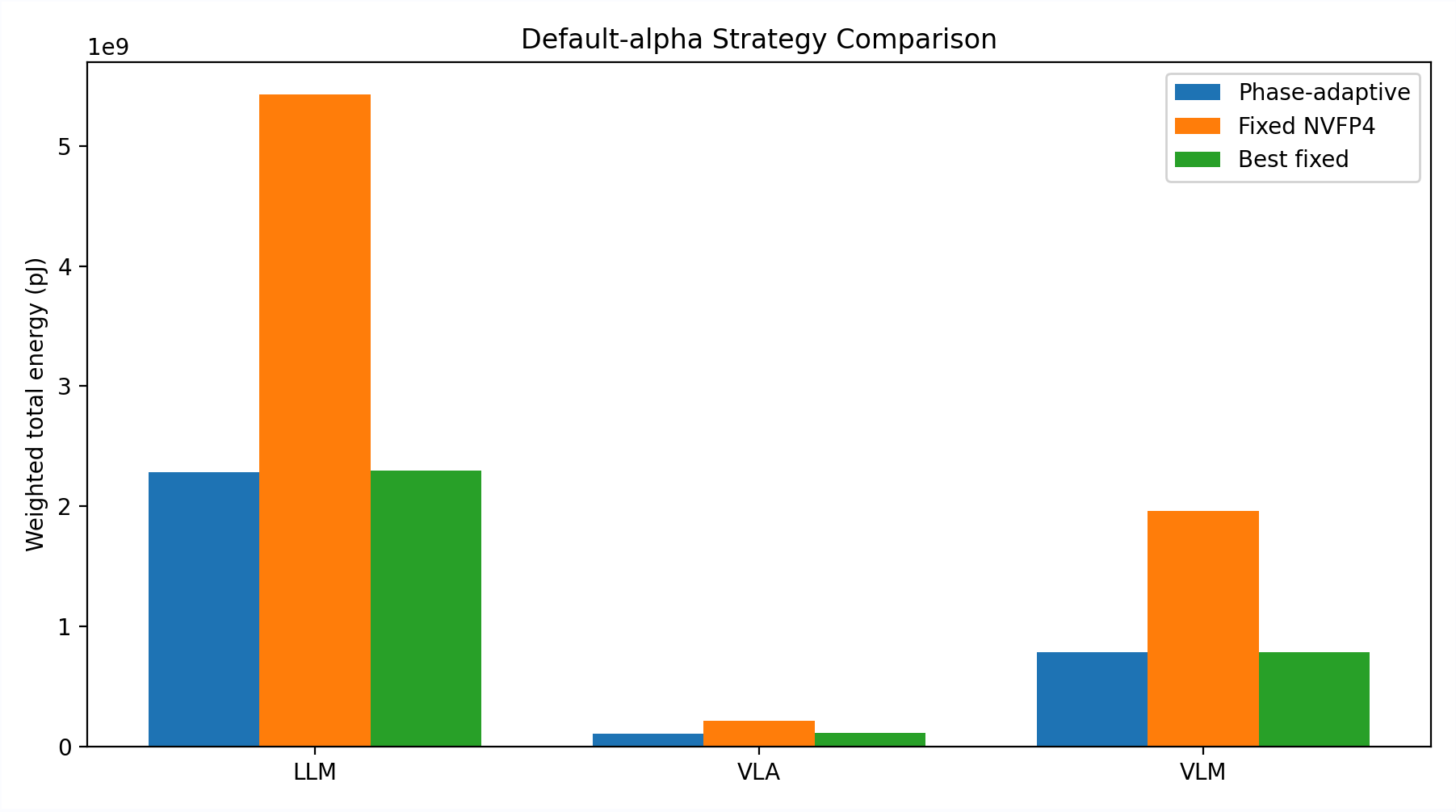
- **LLM**: about 58% lower weighted energy vs C7.
- **VLM**: about 60% lower weighted energy vs C7.
- **VLA**: about 51% lower weighted energy vs C7.

Relative to best-fixed, phase adaptation is selective:

- LLM: about 0.6%
- VLM: 0%
- VLA: about 8.2%



Strategy Comparison at Default α



Phase-adaptive equals best-fixed for VLM because both phases select C1.

Interpretation

C1

MXFP4-LIKE

Often the low-energy knee. Selected for LLM prefill and both VLM phases.

C8

FP16 RESCALE + FP16 ACC

Spends a little more energy to clear accuracy in LLM/VLA decode.

C9

HYBRID TWO-LEVEL

Wins VLA prefill, where a small action-head output dimension changes the tradeoff.

The main contribution is not one universal format; it is the workload- and phase-aware search methodology.

Contributions

1. **Structured search space** for 4-bit MMA rescale pipelines with hardware-realizable datapaths.
2. **Phase-aware evaluation** that separates prefill and decode instead of averaging them away.
3. **Reproducible sweep tooling** that generates AccelForge inputs, runs hardware cost models, emulates accuracy, and joins results.
4. **Completed evidence** that C7/NVFP4-like is consistently high energy, while the optimal choice depends on workload and phase.

Limitations and Next Steps

Current limitations:

- Accuracy uses layer-output cosine on bounded tensor slices, not full end-to-end task metrics.
- AccelForge modeling uses analytical component assumptions rather than synthesized RTL.
- Phase switching overhead is discussed qualitatively; implementation cost needs a tighter area and control estimate.

Next work:

- Larger tensor slices and end-to-end perplexity / task accuracy / success-rate evaluation.
- Milestone-3 architecture saturation analysis across QuantMAC and RescaleMAC counts.
- Final write-up polish and a clearer recommendation for when to enable phase adaptation.

Closing Takeaway

General-purpose 4-bit formats bundle too many choices.

When the accelerator target is known, those choices should be searched:

50-60%

ENERGY REDUCTION VS NVFP4-LIKE
C7

3

DISTINCT SELECTED DATAPATHS

60 + 60

COMPLETED HARDWARE AND
ACCURACY ROWS

Phase-adaptive quantization is useful precisely because it can also tell us when not to adapt.